Module 06 — Vector Databases & Retrieval Operations

> **Goal:** Design, tune and operate the retrieval layer: choose a vector store, tune HNSW, build hybrid (BM25 + dense) search with rank fusion, add cross-encoder reranking, chunk intelligently, and filter by metadata at scale — with the failure modes that only appear in production.

Table of Contents

1. Module Overview & First-Principles Architecture
2. Mathematical Foundations
3. Production-Ready Code
4. Hard-Learned Production Failures
5. Interview Questions

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1. Module Overview & First-Principles Architecture

1.1 Retrieval as a multi-stage funnel

Production retrieval is never "embed query → top-k". It is a funnel that trades recall for precision as candidate count shrinks and per-candidate cost rises:

```
flowchart LR
    Q[User query] --> RW[Query rewrite<br/>HyDE · multi-query · decomposition]
    RW --> D[Dense ANN<br/>HNSW top-100]
    RW --> S[Sparse BM25 / SPLADE<br/>top-100]
    D --> F[Fusion<br/>RRF / weighted]
    S --> F
    F --> M[Metadata & ACL filter]
    M --> R[Cross-encoder rerank<br/>top-50 → top-8]
    R --> P[Context packing<br/>MMR · token budget]
    P --> LLM
```

Stage	Candidates	Cost / candidate	Goal
ANN (dense)	10■–10■ → 100	~µs (index)	High **recall**
BM25 (sparse)	→ 100	~µs	Exact terms, IDs, rare words
Fusion	200 → 100	negligible	Robust blending
Cross-encoder	50–100 → 8	5–30 ms on GPU (batch)	High **precision**

1.2 Vector database landscape

System	Architecture	Index types	Strengths	Watch-outs
Qdrant	Rust, single-binary or distributed (sharding + replicas), segment-based storage with WAL	HNSW (+ payload indexes), scalar/product/binary quantization, sparse vectors, multivector (ColBERT)	Excellent filtered search (filterable HNSW), hybrid via query_points prefetch + fusion, simple ops, on-disk/mmap modes	Rebalancing shards needs planning; cluster feature set evolves
Milvus	Cloud-native disaggregated (proxy, query nodes, data nodes, index nodes; etcd, Pulsar/Kafka, object store)	HNSW, IVF_FLAT/IVF_PQ, DiskANN, GPU CAGRA, SCANN, sparse	Billion-scale; GPU indexes; partitions	Operationally heavy; many components
Pinecone	Managed serverless / pod-based	Proprietary (graph + quantization)	Zero-ops, namespaces for multi-tenancy	Cost at scale; closed internals; data residency
pgvector (Postgres)	Extension; MVCC heap tables	HNSW, IVFFlat; halfvec, bit, sparsevec	Transactional joins with relational data, one system, SQL filters	Single-node scaling limits; HNSW build memory; recall drops with strong filters unless iterative scan
OpenSearch / Elasticsearch	Lucene	HNSW (+ BM25 native)	Hybrid in one engine, mature ops	Heavier JVM footprint

Selection heuristic: < ~5M vectors and you already run Postgres → **pgvector**. Heavy filtering + moderate/large scale + simple ops → **Qdrant**. Billions of vectors / GPU indexing / team with platform skills → **Milvus**. No ops budget → **Pinecone**/managed Qdrant.

1.3 HNSW from first principles

Hierarchical Navigable Small World: a multi-layer proximity graph. Each point is assigned a max layer ■ ~ ■–ln(U)·m_L■ (exponential decay, m_L = 1/ln M). Upper layers are sparse "express lanes"; layer 0 contains all points.

```
Layer 2:  o■■■■■■■■■■■■■■■■■■■o          (few nodes, long links)
Layer 1:  o■■■■■o■■■■■o■■■■■o■■■■■o
Layer 0:  o■o■o■o■o■o■o■o■o■o■o■o■o■o■o  (all nodes, short links, M0 = 2M)
Search: enter at top, greedy descend to nearest in each layer, then beam search (width ef) at layer 0.
```

Parameters: - M (a.k.a. m): max neighbours per node (layers ≥1; 2M at layer 0). ↑M → higher recall, more RAM (~ M·8..16 bytes/vector for links), slower build. - ef_construct: beam width while inserting. ↑ → better graph quality, slower build. - ef (query-time hnsw_ef): beam width at search. ↑ → higher recall, higher latency. **The main runtime knob.**

Memory: RAM ≈ N × (d × bytes_per_dim + M × 2 × 4..8 B) (vectors + links) × ~1.1–1.3 for allocator/overhead.

1.4 Quantization in vector stores

Method	Compression	Recall impact	Notes
Scalar (int8)	4×	–0.5–2 %	Fast; use rescore with original vectors
Product Quantization	8–64×	–3–10 %	Slower distance (LUT); good for huge corpora
Binary (1-bit)	32×	big drop unless d ≥ 1024 and with oversampling+rescore	Works well with high-dim embeddings (OpenAI 3-large, bge-m3)
Matryoshka truncation	d/k	model-dependent	Train with MRL; use shorter prefix for first stage then full for rescoring

Pattern: **quantized vectors in RAM for candidate generation, full-precision on disk (mmap) for rescoring** → large memory reduction at ~1 % recall cost.

1.5 Hybrid search

Dense embeddings capture semantics but fail on exact tokens (SKUs, error codes, names, rare acronyms); BM25 is the reverse. Hybrid fuses both lists. Methods:

1. **Reciprocal Rank Fusion (RRF)** — score-free, robust: score(d) = Σ_i 1/(k + rank_i(d)), k≈60.
2. **Weighted score fusion** — requires score normalisation (min-max or z-score) because BM25 and cosine are on different scales.
3. **Learned sparse (SPLADE, BGE-M3 sparse)** — sparse vectors with term expansion, stored in the same DB.
4. **Late interaction (ColBERT / multivector)** — per-token vectors with MaxSim; higher quality, 10–100× storage.

1.6 Reranking

A **bi-encoder** embeds query and doc independently (fast, index-able) but loses fine-grained interaction. A **cross-encoder** reads [query ; doc] jointly through all transformer layers and outputs relevance — far more accurate but O(candidates) model calls. Standard use: retrieve 50–100, rerank to 5–10. Models: BAAI/bge-reranker-v2-m3, bge-reranker-large, mxbai-rerank, Cohere Rerank 3.x, Jina reranker. LLM listwise rerankers (RankGPT/RankZephyr) are better but slower/costlier.

1.7 Chunking strategies

Strategy	Description	Best for
Fixed-size + overlap	256–512 tokens, 10–20 % overlap	Baseline
Recursive / structure-aware	Split on headings → paragraphs → sentences; keep tables/code intact	Docs, wikis, manuals
Semantic chunking	Split where adjacent-sentence embedding similarity drops	Prose with topic shifts
Parent–child (small-to-big)	Index small chunks (≈128–256 tok) for precision, return parent (≈1,024) for context	Most RAG; best default
Sentence-window	Index sentences; return ±N neighbours	QA over narrative
Late chunking	Embed the whole doc with long-context embedder, pool per chunk	Preserves cross-chunk context
Contextual retrieval	Prepend an LLM-generated 1–2 sentence document context to each chunk before embedding & BM25	Large improvement (~35–49 % fewer retrieval failures in published results) at ingest cost

1.8 Metadata filtering at scale

Filters (tenant_id, acl_groups, doc_type, date) interact badly with ANN: - **Post-filtering** (ANN then filter) can return < k results (or none) when the filter is selective. - **Pre-filtering** (brute force on matching set) is exact but slow for large sets. - **Filterable HNSW** (Qdrant): payload indexes + additional graph links so traversal respects filters; planner switches to brute-force when the filter cardinality is below full_scan_threshold. - Multi-tenancy: one collection + tenant_id payload index (with is_tenant=true optimisation), *not* a collection per tenant beyond a few dozen. Milvus: partition key. Pinecone: namespaces. pgvector: partial indexes or partitioned tables.

---

2. Mathematical Foundations & Formulas

2.1 Similarity metrics

```
cosine(q,d)  = q·d / (||q||||d||)          (normalise once → use dot product)
L2^2(q,d)    = ||q||^2 + ||d||^2 – 2 q·d
dot(q,d)     = q·d                         (use when the model was trained with MIPS)
```

For unit vectors: L2^2 = 2 – 2cos. Use the metric the embedding model was trained for.

2.2 Recall@k and ANN trade-off

```
Recall@k = |ANN_topk ∩ Exact_topk| / k
```

HNSW query cost ≈ O(ef · M · log N) distance computations. Doubling ef roughly doubles latency near the knee while recall rises with diminishing returns (0.90 → 0.97 → 0.99).

2.3 Memory sizing

```
bytes ≈ N × ( d × b_dim + 2·M × 4 ) × 1.2          # float32: b_dim=4; link IDs 4 B
```

Example: N=10M, d=1024, float32, M=16 → 10e6 × (4096 + 128) × 1.2 ≈ 50.7 GB. With int8 scalar quantization in RAM + originals on disk: 10e6 × (1024 + 128) × 1.2 ≈ 13.8 GB.

At 1B vectors, d=768 fp32 = 3 TB raw → need PQ/DiskANN/binary + sharding.

2.4 RRF

```
RRF(d) = Σ_{r ∈ rankers} 1 / (k + rank_r(d)),   k = 60 (default)
```

Properties: ignores raw scores, bounded contribution per ranker; an item ranked 1st by only one retriever (1/61 = 0.0164) loses to one ranked 5th by both (2/65 = 0.0308) — rewards consensus.

2.5 BM25

```
BM25(q,d) = Σ_{t∈q} IDF(t) · ( f(t,d)·(k1+1) ) / ( f(t,d) + k1·(1 – b + b·|d|/avgdl) )
IDF(t) = ln( (N – n_t + 0.5)/(n_t + 0.5) + 1 )
```

Defaults k1=1.2–2.0, b=0.75.

2.6 Ranking metrics

```
MRR      = (1/|Q|) Σ 1/rank_first_relevant
nDCG@k   = DCG@k / IDCG@k,    DCG@k = Σ_{i=1..k} (2^{rel_i} – 1)/log2(i+1)
Hit@k    = fraction of queries with ≥1 relevant in top-k
```

2.7 Retrieval-latency budget

```
T_retrieval = T_embed(q) + max(T_dense, T_sparse) + T_fuse + T_rerank
            ≈ 8 ms + max(15, 10) ms + 1 ms + (n_cand / batch) × t_batch
```

Cross-encoder with 50 candidates × 256 tokens on an L4/A10: ~25–40 ms in one batch. Rerank cost scales linearly with candidates and ~quadratically with sequence length per pair.

2.8 Chunk size vs. recall (empirical shape)

Small chunks → high precision, lost context; large chunks → diluted embeddings (one vector averages many topics). Retrieval quality typically peaks at 200–500 tokens for factoid QA and 500–1,000 for explanatory content; parent–child lets you decouple the *retrieval unit* from the *generation unit*.

2.9 Staleness and index lag

```
freshness_lag = T_source_update → T_visible_in_search
              = T_event + T_chunk + T_embed (batched) + T_upsert + T_index_refresh
```

Qdrant is visible after upsert with wait=true; segment optimisation affects only performance. Elasticsearch default refresh_interval=1s.

---

3. Production-Ready Executable Code

3.1 Local Stack (Qdrant + Redis) via Docker Compose

```
# docker-compose.retrieval.yml
services:
  qdrant:
    image: qdrant/qdrant:v1.12.4
    ports: ["6333:6333", "6334:6334"]
    volumes: ["qdrant-data:/qdrant/storage"]
    environment:
      QDRANT__SERVICE__API_KEY: ${QDRANT_API_KEY:-dev-key}
      QDRANT__STORAGE__PERFORMANCE__MAX_SEARCH_THREADS: "0"
    ulimits: { nofile: { soft: 65535, hard: 65535 } }
    healthcheck:
      test: ["CMD-SHELL", "bash -c ':> /dev/tcp/127.0.0.1/6333' || exit 1"]
      interval: 10s
      retries: 10
  tei-embed:
    image: ghcr.io/huggingface/text-embeddings-inference:1.5
    command: --model-id BAAI/bge-m3 --max-batch-tokens 65536
    ports: ["8081:80"]
    deploy: { resources: { reservations: { devices: [{ driver: nvidia, count: 1, capabilities: [gpu] }] } }
  } }
  tei-rerank:
    image: ghcr.io/huggingface/text-embeddings-inference:1.5
    command: --model-id BAAI/bge-reranker-v2-m3
    ports: ["8082:80"]
    deploy: { resources: { reservations: { devices: [{ driver: nvidia, count: 1, capabilities: [gpu] }] } }
  } }
volumes:
  qdrant-data:
```

3.2 Collection Design: Dense + Sparse + Payload Indexes + Quantization + Multi-tenancy

```
"""Create a production Qdrant collection.""&q
uot;
from __future__ import annotations

from qdrant_client import QdrantClient, models

client = QdrantClient(url="http://localhost:6333", api_key="dev-key", timeout=30, pre
fer_grpc=True)
COLL = "kb_v3"

if client.collection_exists(COLL):
    raise SystemExit(f"{COLL} exists; create a new versioned collection and swap the alias")

client.create_collection(
    collection_name=COLL,
    vectors_config={
        "dense": models.VectorParams(
            size=1024,
            distance=models.Distance.COSINE,
            hnsw_config=models.HnswConfigDiff(m=16, ef_construct=200, on_disk=False),
            on_disk=True,                    # originals on disk (mmap) for rescoring
        )
```

```
    },
    sparse_vectors_config={"sparse": models.SparseVectorParams(index=models.SparseIndexParams(o
n_disk=False))},
    quantization_config=models.ScalarQuantization(
        scalar=models.ScalarQuantizationConfig(type=models.ScalarType.INT8, quantile=0.99, always_ram=Tru
e)
    ),
    optimizers_config=models.OptimizersConfigDiff(indexing_threshold=20_000, default_segment_number=4),
    shard_number=2,
    replication_factor=2,
    on_disk_payload=True,
)

# Payload indexes are mandatory for filtered search performance.
client.create_payload_index(COLL, "tenant_id", models.KeywordIndexParams(type="keyword&quo
t;, is_tenant=True))
client.create_payload_index(COLL, "acl_groups", models.PayloadSchemaType.KEYWORD)
client.create_payload_index(COLL, "doc_type", models.PayloadSchemaType.KEYWORD)
client.create_payload_index(COLL, "updated_at", models.PayloadSchemaType.DATETIME)

# Alias = atomic blue/green swap for re-indexing or embedding-model upgrades.
client.update_collection_aliases(
    change_aliases_operations=[models.CreateAliasOperation(create_alias=models.CreateAlias(collection_nam
e=COLL, alias_name="kb_current"))]
)
```

3.3 Ingestion Pipeline: Parent–Child Chunking + Contextual Enrichment + Batched Embedding

```
"""Ingest documents: structure-aware parent/child chunk
s, embeddings (dense+sparse), idempotent upserts."""
from __future__ import annotations

import hashlib
import re
import uuid
from dataclasses import dataclass
from typing import Iterable, Iterator, Sequence

import httpx
from qdrant_client import QdrantClient, models
from tenacity import retry, stop_after_attempt, wait_exponential

TEI = "http://localhost:8081"
client = QdrantClient(url="http://localhost:6333", api_key="dev-key", prefer_grpc=Tru
e)
_HEADING = re.compile(r"^(#{1,4})\s+(.*)$", re.M)


@dataclass(frozen=True)
class Chunk:
    id: str
    parent_id: str
    text: str
    parent_text: str
    doc_id: str
    section: str
    tenant_id: str
    acl_groups: tuple[str, ...]
    doc_type: str
    updated_at: str


def approx_tokens(s: str) -> int:
```

```
    return max(1, len(s) // 4)


def split_sections(md: str) -> Iterator[tuple[str, str]]:
    """Yield (heading_path, body) preserving heading hierarchy."""
    stack: list[tuple[int, str]] = []
    last = 0
    heads = list(_HEADING.finditer(md))
    if not heads:
        yield "", md
        return
    if heads[0].start() > 0:
        yield "", md[: heads[0].start()]
    for i, m in enumerate(heads):
        lvl, title = len(m.group(1)), m.group(2).strip()
        while stack and stack[-1][0] >= lvl:
            stack.pop()
        stack.append((lvl, title))
        end = heads[i + 1].start() if i + 1 < len(heads) else len(md)
        yield " > ".join(t for _, t in stack), md[m.end():end].strip()


def sentences(text: str) -> list[str]:
    return [s.strip() for s in re.split(r"(?<=[.!?])\s+|\n{2,}", text) if s.strip()]


def pack(sents: Sequence[str], max_tok: int, overlap_sents: int = 1) -> Iterator[str]:
    buf: list[str] = []
    tok = 0
    for s in sents:
        t = approx_tokens(s)
        if buf and tok + t > max_tok:
            yield " ".join(buf)
            buf = buf[-overlap_sents:] if overlap_sents else []
            tok = sum(approx_tokens(x) for x in buf)
        buf.append(s)
        tok += t
    if buf:
        yield " ".join(buf)


def make_chunks(doc_id: str, md: str, *, tenant_id: str, acl: tuple[str, ...], doc_type: str, updated_at:
 str,
                child_tok: int = 200, parent_tok: int = 900) -> Iterator[Chunk]:
    for section, body in split_sections(md):
        for parent in pack(sentences(body), parent_tok, overlap_sents=0):
            pid = hashlib.sha1(f"{doc_id}|{section}|{parent[:64]}".encode()).hexdigest()[:16]
            for child in pack(sentences(parent), child_tok, overlap_sents=1):
                cid = str(uuid.uuid5(uuid.NAMESPACE_URL, f"{doc_id}|{pid}|{child}"))  # idempot
ent
                yield Chunk(cid, pid, child, parent, doc_id, section, tenant_id, acl, doc_type, updated_a
t)


@retry(stop=stop_after_attempt(5), wait=wait_exponential(min=0.5, max=10))
def embed(texts: list[str]) -> list[list[float]]:
    r = httpx.post(f"{TEI}/embed", json={"inputs": texts, "normalize": True
, "truncate": True}, timeout=60)
    r.raise_for_status()
    return r.json()


def bm25_sparse(text: str) -> models.SparseVector:
    """Lightweight lexical sparse vector (hashing TF). Swap for fastembed SPLADE/BM25 in p
roduction."""
    counts: dict[int, float] = {}
```

```
    for tok in re.findall(r"[a-z0-9_]+", text.lower()):
        h = int(hashlib.md5(tok.encode()).hexdigest()[:8], 16) % (1 << 20)
        counts[h] = counts.get(h, 0.0) + 1.0
    idx = sorted(counts)
    return models.SparseVector(indices=idx, values=[1.0 + __import__("math").log(counts[i]) for
 i in idx])


def upsert(chunks: Iterable[Chunk], collection: str = "kb_current", batch: int = 64) -> int:
    n = 0
    buf: list[Chunk] = []

    def flush() -> None:
        nonlocal n
        if not buf:
            return
        dense = embed([f"{c.section}\n{c.text}" for c in buf])  # section heading = cheap cont
extual retrieval
        points = [
            models.PointStruct(
                id=c.id,
                vector={"dense": d, "sparse": bm25_sparse(f"{c.section} {c.text}
")},
                payload={"text": c.text, "parent_id": c.parent_id, "parent_text&
quot;: c.parent_text, "doc_id": c.doc_id,
                         "section": c.section, "tenant_id": c.tenant_id, "acl_gr
oups": list(c.acl_groups),
                         "doc_type": c.doc_type, "updated_at": c.updated_at},
            )
            for c, d in zip(buf, dense)
        ]
        client.upsert(collection, points=points, wait=True)
        n += len(points)
        buf.clear()

    for c in chunks:
        buf.append(c)
        if len(buf) >= batch:
            flush()
    flush()
    return n


def delete_doc(doc_id: str, collection: str = "kb_current") -> None:
    """Re-ingest = delete by doc_id then upsert; prevents orphaned chunks when a doc shrin
ks."""
    client.delete(collection, points_selector=models.FilterSelector(
        filter=models.Filter(must=[models.FieldCondition(key="doc_id", match=models.MatchValue(
value=doc_id))])))
```

3.4 Hybrid Search (server-side RRF fusion) with ACL-aware Filtering

```
"""Single round-trip hybrid retrieval: dense + sparse p
refetch fused by RRF inside Qdrant."""
from __future__ import annotations

from dataclasses import dataclass

from qdrant_client import QdrantClient, models

client = QdrantClient(url="http://localhost:6333", api_key="dev-key", prefer_grpc=Tru
e)


@dataclass(frozen=True)
```

```
class Hit:
    id: str
    score: float
    text: str
    parent_text: str
    doc_id: str
    section: str


def build_filter(tenant_id: str, user_groups: list[str], doc_types: list[str] | None, since_iso: str | No
ne) -> models.Filter:
    must: list[models.Condition] = [
        models.FieldCondition(key="tenant_id", match=models.MatchValue(value=tenant_id)),
        # ACL: the chunk is visible if ANY of its groups intersects the user's groups.
        models.FieldCondition(key="acl_groups", match=models.MatchAny(any=user_groups + ["
public"])),
    ]
    if doc_types:
        must.append(models.FieldCondition(key="doc_type", match=models.MatchAny(any=doc_types))
)
    if since_iso:
        must.append(models.FieldCondition(key="updated_at", range=models.DatetimeRange(gte=sinc
e_iso)))
    return models.Filter(must=must)


def hybrid_search(q_dense: list[float], q_sparse: models.SparseVector, flt: models.Filter, *, limit: int
= 50,
                  prefetch: int = 100, hnsw_ef: int = 128) -> list[Hit]:
    res = client.query_points(
        collection_name="kb_current",
        prefetch=[
            models.Prefetch(query=q_dense, using="dense", limit=prefetch, filter=flt,
                            params=models.SearchParams(hnsw_ef=hnsw_ef,
                                                       quantization=models.QuantizationSearchParams(resco
re=True, oversampling=2.0))),
            models.Prefetch(query=q_sparse, using="sparse", limit=prefetch, filter=flt),
        ],
        query=models.FusionQuery(fusion=models.Fusion.RRF),
        limit=limit,
        with_payload=True,
    )
    return [Hit(str(p.id), p.score, p.payload["text"], p.payload["parent_text"], p.pa
yload["doc_id"], p.payload["section"])
            for p in res.points]
```

3.5 Cross-Encoder Reranking (local TEI or sentence-transformers) + Cohere option

```
"""Rerank with BGE reranker via TEI; fallback to score
order on failure (never fail the request)."""
from __future__ import annotations

import logging
import time
from typing import Sequence

import httpx

log = logging.getLogger("rerank")
RERANK_URL = "http://localhost:8082/rerank"


def rerank(query: str, hits: Sequence["Hit"], top_n: int = 8, timeout_s: float = 0.4) -> lis
t["Hit"]:
    if not hits:
```

```
        return []
    t0 = time.perf_counter()
    try:
        r = httpx.post(RERANK_URL, json={"query": query, "texts": [h.text for h in hi
ts], "truncate": True}, timeout=timeout_s)
        r.raise_for_status()
        order = sorted(r.json(), key=lambda x: x["score"], reverse=True)[:top_n]
        out = [hits[o["index"]] for o in order]
        log.info("rerank ok n=%d ms=%.1f", len(hits), (time.perf_counter() - t0) * 1e3)
        return out
    except Exception as e:          # degrade gracefully to fused order
        log.warning("rerank failed (%s); falling back to fusion order", e)
        return list(hits[:top_n])


# Cohere variant:
#   import cohere; co = cohere.ClientV2()
#   r = co.rerank(model="rerank-v3.5", query=q, documents=[h.text for h in hits], top_n=8)
#   out = [hits[x.index] for x in r.results]
```

Parent-document expansion and MMR packing:

```
import numpy as np

def dedupe_parents(hits: list["Hit"], max_parents: int = 5) -> list[str]:
    seen: set[str] = set(); out: list[str] = []
    for h in hits:
        key = h.doc_id + "|" + h.parent_text[:60]
        if key in seen:
            continue
        seen.add(key); out.append(h.parent_text)
        if len(out) == max_parents:
            break
    return out

def mmr(query_vec: np.ndarray, doc_vecs: np.ndarray, k: int, lam: float = 0.7) -> list[int]:
    sims = doc_vecs @ query_vec
    chosen: list[int] = []; cand = list(range(len(doc_vecs)))
    while cand and len(chosen) < k:
        scores = [lam * sims[i] - (1 - lam) * max((doc_vecs[i] @ doc_vecs[j] for j in chosen), default=0.
0) for i in cand]
        best = cand[int(np.argmax(scores))]
        chosen.append(best); cand.remove(best)
    return chosen
```

3.6 HNSW Tuning Harness (measure recall vs latency, choose ef)

```
"""Sweep hnsw_ef; compare against exact (brute-force) r
esults to compute recall@k."""
from __future__ import annotations

import statistics
import time

import numpy as np
from qdrant_client import QdrantClient, models

client = QdrantClient(url="http://localhost:6333", api_key="dev-key", prefer_grpc=Tru
e)
COLL = "kb_current"


def sweep(queries: list[list[float]], k: int = 10, efs: tuple[int, ...] = (16, 32, 64, 128, 256, 512)) -&
gt; None:
```

```
    exact = [
        {p.id for p in client.query_points(COLL, query=q, using="dense", limit=k,
                                           search_params=models.SearchParams(exact=True)).points}
        for q in queries
    ]
    print(f"{'ef':>5} {'recall@k':>9} {'p50 ms':>8} {'p95 ms':>8}")
    for ef in efs:
        lat: list[float] = []; rec: list[float] = []
        for q, truth in zip(queries, exact):
            t = time.perf_counter()
            got = {p.id for p in client.query_points(COLL, query=q, using="dense", limit=k,
                                                     search_params=models.SearchParams(hnsw_ef=ef)).point
s}
            lat.append((time.perf_counter() - t) * 1e3)
            rec.append(len(got & truth) / k)
        print(f"{ef:>5} {np.mean(rec):>9.3f} {statistics.median(lat):>8.1f} {np.percentile(
lat, 95):>8.1f}")
```

Typical guidance: pick the smallest ef giving recall@10 ≥ 0.97–0.98 at p95 latency within budget; rebuild with higher m/ef_construct only if no ef reaches target.

3.7 Retrieval Evaluation (Hit@k / MRR / nDCG) with a Golden Set

```
from __future__ import annotations

import math


def hit_at_k(results: list[list[str]], gold: list[set[str]], k: int) -> float:
    return sum(1 for r, g in zip(results, gold) if set(r[:k]) & g) / len(gold)


def mrr(results: list[list[str]], gold: list[set[str]]) -> float:
    total = 0.0
    for r, g in zip(results, gold):
        for i, d in enumerate(r, 1):
            if d in g:
                total += 1 / i
                break
    return total / len(gold)


def ndcg_at_k(results: list[list[str]], gold: list[dict[str, int]], k: int) -> float:
    def dcg(rels: list[int]) -> float:
        return sum((2**rel - 1) / math.log2(i + 2) for i, rel in enumerate(rels))
    vals = []
    for r, g in zip(results, gold):
        rels = [g.get(d, 0) for d in r[:k]]
        ideal = sorted(g.values(), reverse=True)[:k]
        vals.append(dcg(rels) / dcg(ideal) if ideal else 0.0)
    return sum(vals) / len(vals)
```

CI gate example: Hit@10 ≥ 0.92, MRR ≥ 0.55, nDCG@10 ≥ 0.70, and *no query-class slice* more than 5 points under the mean.

3.8 pgvector Option (SQL, HNSW, filtered iterative scan)

```
CREATE EXTENSION IF NOT EXISTS vector;

CREATE TABLE chunks (
  id          uuid PRIMARY KEY,
  tenant_id   text NOT NULL,
  acl_groups  text[] NOT NULL,
  doc_id      text NOT NULL,
```

```
  content     text NOT NULL,
  embedding   halfvec(1024) NOT NULL,      -- half precision halves memory
  tsv         tsvector GENERATED ALWAYS AS (to_tsvector('english', content)) STORED,
  updated_at  timestamptz NOT NULL DEFAULT now()
);

CREATE INDEX chunks_hnsw ON chunks USING hnsw (embedding halfvec_cosine_ops) WITH (m = 16, ef_constructio
n = 200);
CREATE INDEX chunks_gin  ON chunks USING gin (tsv);
CREATE INDEX chunks_tenant ON chunks (tenant_id);

-- Filtered ANN: enable iterative scan so selective filters still return k rows (pgvector >= 0.8)
SET hnsw.ef_search = 100;
SET hnsw.iterative_scan = relaxed_order;
SET hnsw.max_scan_tuples = 20000;

-- Hybrid in one query via RRF
WITH dense AS (
  SELECT id, ROW_NUMBER() OVER (ORDER BY embedding <=> $1) AS r
  FROM chunks WHERE tenant_id = $2 AND acl_groups && $3
  ORDER BY embedding <=> $1 LIMIT 100),
sparse AS (
  SELECT id, ROW_NUMBER() OVER (ORDER BY ts_rank_cd(tsv, q) DESC) AS r
  FROM chunks, websearch_to_tsquery('english', $4) q
  WHERE tenant_id = $2 AND acl_groups && $3 AND tsv @@ q
  ORDER BY ts_rank_cd(tsv, q) DESC LIMIT 100)
SELECT COALESCE(d.id, s.id) AS id,
       COALESCE(1.0/(60+d.r), 0) + COALESCE(1.0/(60+s.r), 0) AS rrf
FROM dense d FULL OUTER JOIN sparse s USING (id)
ORDER BY rrf DESC LIMIT 50;
```

3.9 Milvus Option (partition key multi-tenancy + hybrid + RRF)

```
from __future__ import annotations

from pymilvus import (AnnSearchRequest, Collection, CollectionSchema, DataType, FieldSchema, RRFRanker, c
onnections, utility)

connections.connect(uri="http://localhost:19530")
fields = [
    FieldSchema("id", DataType.VARCHAR, is_primary=True, max_length=64),
    FieldSchema("tenant_id", DataType.VARCHAR, max_length=64, is_partition_key=True),
    FieldSchema("text", DataType.VARCHAR, max_length=8192),
    FieldSchema("dense", DataType.FLOAT_VECTOR, dim=1024),
    FieldSchema("sparse", DataType.SPARSE_FLOAT_VECTOR),
]
coll = Collection("kb_v3", CollectionSchema(fields), num_partitions=64)
coll.create_index("dense", {"index_type": "HNSW", "metric_type":
"COSINE", "params": {"M": 16, "efConstruction": 200}})
coll.create_index("sparse", {"index_type": "SPARSE_INVERTED_INDEX", "m
etric_type": "IP"})
coll.load()


def hybrid(q_dense: list[float], q_sparse: dict[int, float], tenant: str, k: int = 50):
    expr = f'tenant_id == "{tenant}"'
    reqs = [
        AnnSearchRequest([q_dense], "dense", {"metric_type": "COSINE", &quo
t;params": {"ef": 128}}, limit=100, expr=expr),
        AnnSearchRequest([q_sparse], "sparse", {"metric_type": "IP"}, limit
=100, expr=expr),
    ]
    return coll.hybrid_search(reqs, RRFRanker(60), limit=k, output_fields=["text"])
```

3.10 Embedding-Model Migration (zero-downtime)

```
"""Blue/green re-index when the embedding model changes
 (vectors from different models are NOT comparable)."""
from __future__ import annotations

from qdrant_client import QdrantClient, models

client = QdrantClient(url="http://localhost:6333", api_key="dev-key")


def migrate(old: str, new: str, alias: str, reembed, batch: int = 256) -> None:
    offset = None
    while True:
        pts, offset = client.scroll(old, limit=batch, offset=offset, with_payload=True, with_vectors=Fals
e)
        if not pts:
            break
        vecs = reembed([p.payload["text"] for p in pts])          # new model
        client.upsert(new, [models.PointStruct(id=p.id, vector={"dense": v}, payload=p.payload)
 for p, v in zip(pts, vecs)])
        if offset is None:
            break
    # 1) shadow-evaluate new vs old on the golden set  2) flip alias atomically
    client.update_collection_aliases([
        models.DeleteAliasOperation(delete_alias=models.DeleteAlias(alias_name=alias)),
        models.CreateAliasOperation(create_alias=models.CreateAlias(collection_name=new, alias_name=alias
)),
    ])
```

3.11 Kubernetes: Qdrant Cluster via StatefulSet (core settings)

```
apiVersion: apps/v1
kind: StatefulSet
metadata: { name: qdrant }
spec:
  serviceName: qdrant-headless
  replicas: 3
  selector: { matchLabels: { app: qdrant } }
  template:
    metadata: { labels: { app: qdrant } }
    spec:
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector: { matchLabels: { app: qdrant } }
              topologyKey: topology.kubernetes.io/zone
      containers:
        - name: qdrant
          image: qdrant/qdrant:v1.12.4
          args: ["./entrypoint.sh"]
          env:
            - { name: QDRANT__CLUSTER__ENABLED, value: "true" }
            - { name: QDRANT__SERVICE__API_KEY, valueFrom: { secretKeyRef: { name: qdrant, key: api-key }
} }
          ports: [{ containerPort: 6333 }, { containerPort: 6334 }, { containerPort: 6335 }]
          resources: { requests: { cpu: "4", memory: 24Gi }, limits: { memory: 32Gi } }
          readinessProbe: { httpGet: { path: /readyz, port: 6333 }, periodSeconds: 5 }
          volumeMounts: [{ name: storage, mountPath: /qdrant/storage }]
  volumeClaimTemplates:
    - metadata: { name: storage }
      spec: { accessModes: [ReadWriteOnce], storageClassName: fast-ssd, resources: { requests: { storage:
500Gi } } }
```

4. Hard-Learned Production Failures & Edge Cases

Incident 1 — Filtered search returned zero results for a small tenant

Symptom. Tenants with few documents saw "I can't find anything" while the content existed; large tenants were fine.

Root-cause analysis. 1. The system did ANN top-100 globally, then applied tenant_id filter in application code (post-filtering). 2. A small tenant owned 0.01 % of vectors; the global top-100 contained none of their chunks. 3. Switching to server-side filtering without a payload index made Qdrant fall back to scanning the payload — latency of 4 s at 20M points.

Remediation. Create a keyword payload index with is_tenant=true (co-locates tenant vectors), pass the filter to Qdrant so the planner uses filterable HNSW or brute-force when cardinality is low, monitor filter_cardinality and p95 latency per tenant size bucket. In pgvector enable hnsw.iterative_scan.

Incident 2 — Stale or deleted documents kept being cited (and a permission leak)

Symptom. A user quoted a policy that was withdrawn six months earlier; separately, an employee saw a snippet from a document they had lost access to.

Root-cause analysis. 1. Ingestion only upserted; when a document shrank or was deleted, orphaned chunks remained (the old chunk IDs were never removed). 2. ACLs were copied into payload at ingest time and never refreshed when group membership/permissions changed.

Remediation. - Idempotent re-ingestion: delete_doc(doc_id) then upsert, or tombstone by doc_version and filter doc_version == current. - Event-driven permission sync (webhook from IdP/DMS → set_payload on affected doc_id), with a nightly reconciliation job comparing source ACLs to payload ACLs and alerting on drift. - **Enforce authorization at query time** with a filter built from the *current* user token groups (never trust cached ACLs alone) and re-check against the source of truth for top-k before returning snippets for high-sensitivity corpora. - Add valid_until payload field and filter on it.

Incident 3 — Embedding model upgrade silently destroyed recall

Symptom. After swapping bge-large → bge-m3 Hit@10 dropped from 0.91 to 0.31 within one hour; no errors.

Root-cause analysis. New documents were embedded with the new model while the existing 12M vectors came from the old one, in the same collection. Cosine between spaces is meaningless; queries (embedded with the new model) only matched new documents.

Remediation. Never mix embedding models in a collection. Use the blue/green migration of §3.10, store embedding_model and embedding_version in the collection name and payload, and add a startup assertion that the query embedder's model id equals the collection's metadata. Shadow-evaluate with the golden set before flipping the alias.

Incident 4 — HNSW memory blow-up on index build / OOM kills

Symptom. Qdrant pods OOM-killed during a bulk load of 60M vectors; restarts looped.

Root cause. ef_construct=512, m=64, all vectors in RAM, indexing_threshold low → many concurrent HNSW builds, and the working-set exceeded the pod limit (memory limit = request, no mmap headroom).

Remediation. Bulk-load with indexing disabled (indexing_threshold=0), then enable indexing; use scalar quantization with always_ram, originals on_disk; m=16, ef_construct=128–200; size memory with §2.3 plus 30 % headroom; stagger ingestion workers; set a higher memory limit than request for page-cache.

Incident 5 — Reranker latency tail wrecked the SLA

Symptom. p99 retrieval latency 3 s, though median was 90 ms.

Root cause. Reranker received 100 candidates of variable length (some 4k-token chunks) in one batch; a single long pair dominated padded batch compute; GPU shared with embedding service.

Remediation. Truncate pairs (max_length=512), cap candidates at 50, dynamic batching by length, separate GPU/pool for reranker, a 400 ms timeout with graceful fallback to fusion order (§3.5), and cache rerank scores for hot queries.

Incident 6 — "Lost in the middle": more context lowered answer quality

Symptom. Raising top-k from 5 to 20 reduced faithfulness.

Root cause. Long contexts bury the relevant passage mid-prompt; models attend better to start/end. Also the added chunks included distractors.

Remediation. Rerank, keep k small (5–8), place highest-ranked at the start and end, use parent-document expansion only for the top few, and evaluate the context-precision metric (Module 07) when changing k.

---

5. Senior/Staff LLMOps Interview Questions

Scenario 1 — "Dense retrieval misses queries containing product SKUs and error codes. Fix it."

Junior/Mid Trap Answer: "Use a bigger/better embedding model or fine-tune the embedder."

Staff Production Answer: "Embedding models compress semantics; arbitrary identifiers like ERR_4012 or SKU-88-AX are out-of-vocabulary-ish and get fuzzy vectors. The right tool is lexical matching, so I'd implement **hybrid search**: dense ANN plus BM25 or learned-sparse (SPLADE/BGE-M3 sparse), fused with RRF (k≈60) because it needs no score normalisation, then cross-encoder rerank the top 50. I'd also tokenise identifiers sensibly for BM25 (preserve punctuation, n-gram analyser), maybe add a query classifier that boosts the sparse leg when the query contains code-like tokens, and measure the improvement on a golden set sliced by query type (identifier vs natural-language). Fine-tuning the embedder is a last resort and risks regression elsewhere."

Scenario 2 — "Choose a vector database for 200M chunks, multi-tenant, strict ACL filtering, 100 QPS, p95 < 100 ms."

Junior/Mid Trap Answer: "Pinecone because it's managed" (or "pgvector because we already have Postgres") without analysis.

Staff Production Answer: "Size first: 200M × 1024-d float32 = 820 GB raw; with HNSW links ~1 TB. That rules out single-node pgvector. With int8 scalar quantization in RAM (~230 GB) and originals on NVMe for rescoring, it fits a 6–8 node cluster. Because ACL and tenant filters are on every query, the engine must support **filterable HNSW** — Qdrant is a strong fit (payload indexes, is_tenant, planner switching to brute-force for selective filters, hybrid prefetch+RRF); Milvus with partition keys is the alternative if we also need GPU indexing or higher scale. I'd shard by tenant hash with 2× replication across zones, keep 99 % recall via ef tuning, and load-test with the real filter-selectivity distribution, not unfiltered queries. Operational plan: blue/green collections via aliases, ACL sync pipeline, per-tenant quotas. Managed Pinecone is viable if ops capacity is the constraint, but I'd model cost at 200M vectors and check data-residency/ACL-filter semantics."

Scenario 3 — "Why add a reranker if the embedding model is already good?"

Junior/Mid Trap Answer: "It improves accuracy a bit."

Staff Production Answer: "Bi-encoders compress each document into one vector independently, so they cannot model query–document token interactions; they optimise for recall over a huge corpus cheaply. A cross-encoder attends jointly over query and passage and is far more precise, but costs a forward pass per pair, so it's only affordable on a short candidate list. The funnel is: ANN/hybrid top-100 for recall → cross-encoder top-5–8 for precision. Typical gains: +10–25 % nDCG@10 and, importantly, higher *context precision*, which improves groundedness because fewer distractors reach the LLM. Costs: ~20–40 ms GPU for 50 pairs, an extra service to operate. I'd add timeouts with fallback to fusion order, cap candidates and sequence length, and verify the gain on the golden set — if nDCG doesn't move, skip it."

Scenario 4 — "Walk me through chunking for a corpus of PDFs with tables, code blocks and long policies."

Junior/Mid Trap Answer: "Split every 500 characters with 50 overlap."

Staff Production Answer: "Fixed character splitting shreds tables and code and strips headings. I'd parse structure first (layout-aware parser such as Docling/Unstructured; tables as Markdown/HTML units; code fenced), then split recursively on headings → paragraphs → sentences. I'd use **parent–child**: embed ~200-token child chunks for precision but return the ~900-token parent (or full section) for generation. Each chunk's embedding text is prefixed with its heading path (and optionally an LLM-generated contextual blurb — Anthropic-style contextual retrieval) so 'Section 4.2: Termination' is not lost. Tables are kept whole or summarised with the raw table in payload. Store doc_id, section, page, version, acl. Then tune empirically: sweep chunk

sizes (128/256/512) and measure Hit@k and downstream faithfulness; there's no universal best size."

Scenario 5 — "Recall dropped after we added a tenant filter. Why and how to fix?"

Junior/Mid Trap Answer: "The filter excludes some results, so recall is naturally lower."

Staff Production Answer: "That conflates correct filtering with ANN failure. If the engine post-filters the ANN candidates, a selective filter eliminates most of the top-ef candidates and you get fewer than k results — a recall artefact. If it pre-filters the graph without filter-aware links, HNSW traversal can get disconnected. Remedies: use an engine with filter-aware HNSW (Qdrant builds extra links per payload index, switches to exact scan under a cardinality threshold), create payload indexes on every filtered field, raise ef for filtered queries, or use iterative scans (pgvector ≥0.8). For hard multi-tenancy, partition (Milvus partition key, Pinecone namespaces, Qdrant tenant index) so each tenant's graph is separate. I'd measure recall against an exact filtered scan per selectivity bucket (1 %, 0.1 %, 0.01 %) to find the degraded regime."

Scenario 6 — "How do you keep the index fresh and consistent with the source of truth?"

Junior/Mid Trap Answer: "Re-run the full ingestion nightly."

Staff Production Answer: "Nightly full rebuilds create up-to-24 h staleness and a large compute bill. I'd use CDC/webhooks from source systems onto a queue (Kafka/SQS), idempotent workers keyed by doc_id + content_hash that skip unchanged documents, delete-then-upsert per document (so shrunk docs leave no orphans) or version tag + filter on current version, and separate pipelines for permission changes (cheap set_payload) vs content changes (re-chunk, re-embed). A reconciliation job compares source inventory with index inventory and alerts on drift (count mismatch, missing doc, extra doc). Track freshness SLO as now – source_updated_at for the p95 of documents. Embedding model upgrades are blue/green into a new collection with alias flip. Deletions for compliance (GDPR) need guaranteed propagation and verification queries."

Scenario 7 — "Explain HNSW parameters and how you'd tune them."

Junior/Mid Trap Answer: "Higher values are always more accurate so set them as high as possible."

Staff Production Answer: "M controls graph degree (memory ~ 8·M bytes per vector for links, plus build time); ef_construct is the build-time beam width (index quality vs build cost); ef/hnsw_ef is the query-time beam width (the main recall/latency dial). Maxing them wastes RAM and CPU with diminishing returns. My procedure: fix M=16, ef_construct=128–200, build, then sweep ef against an exact-search ground truth on ~1k representative queries to chart recall@k vs p95 latency; choose the smallest ef meeting recall ≥0.97 within the latency budget. Only if no ef reaches target do I rebuild with higher M/ef_construct. For memory: scalar quantization with rescoring recovers most recall at 4× less RAM. Re-tune when data distribution, dimensionality or filters change."

Scenario 8 — "Users ask questions that need multiple documents (comparisons, multi-hop). Single-shot retrieval fails. What do you do?"

Junior/Mid Trap Answer: "Increase top-k so more documents are retrieved."

Staff Production Answer: "More k adds noise rather than the missing hop. I'd decompose: an LLM-based query planner splits into sub-questions (or generates step-back/multi-query variants), retrieves per sub-question, fuses and reranks per leg, then synthesises with citations. For true multi-hop (answer to hop 1 is a query term for hop 2) use an iterative/agentic loop with a max-hops cap and a stop criterion, or GraphRAG for entity-relationship-heavy corpora (pre-extracted knowledge graph + community summaries). Cost grows with each hop, so route: a cheap classifier sends simple queries down the single-shot path. Evaluate with multi-hop sets (HotpotQA-style internal data) and track latency/cost per route."

Key Takeaways

- Retrieval is a **funnel**: recall first (hybrid ANN + BM25), precision second (cross-encoder), packing last.
- Filters need payload indexes and filter-aware ANN; test recall per filter-selectivity bucket.
- Never mix embedding models in one index; migrate blue/green with aliases.
- Parent–child chunking + heading-prefixed embeddings is the best default; measure, don't guess.
- Authorization belongs in the query path, not just the ingestion path; keep freshness and ACLs event-driven.

6.6 Graph and Multi-Hop Retrieval Operations (GraphRAG)

Conceptual Deep Dive

Adopt graph retrieval only after measuring that hybrid retrieval fails on relational, multi-hop or corpus-wide questions ("which suppliers of X also operate in region Y?", "what are the main themes?"). If BM25 plus dense retrieval answers your FAQ-style traffic, a graph adds cost and failure modes without recall.

The canonical **GraphRAG** pipeline has five stages: (1) an LLM extracts entities and relations from each chunk; (2) **entity resolution** merges aliases and normalised names into canonical nodes; (3) community detection (Leiden) clusters the graph hierarchically; (4) an LLM writes a summary per community; (5) queries run in **local** mode (find entities near the query, expand a few hops, return their source chunks) or **global** mode (map-reduce over community summaries). Global search answers questions no single chunk or neighbourhood contains, such as "what are the recurring risks across all contracts?". Lighter variants often capture most of the value: entity-linked chunk graphs, document link graphs (citations, hyperlinks, ticket references) and hierarchical summaries.

Cost is dominated by indexing: roughly one or more LLM calls per chunk for extraction, plus summary generation, and extraction errors compound (a wrong edge is retrieved confidently). Updates are expensive: prefer incremental strategies (re-extract only changed chunks, tombstone edges whose source documents were deleted, recompute only affected communities on a schedule) over full rebuilds. Contradictory relations from different sources should keep both with provenance and timestamps instead of silently picking one; stale edges after a document deletion are a correctness bug, so edges must carry source document ids.

Storage choice is operational: a graph database gives traversal and Cypher ergonomics; adjacency tables in Postgres (edge table with source, relation, target, doc id, ACL) are often enough for two or three hops and reuse existing backup and access control. Never execute unconstrained LLM-generated Cypher or SQL against production; use templated queries with validated parameters, read-only roles, timeouts and row limits.

Security is the sharp edge. Derived artefacts (summaries, edges, community reports) mix information from many documents and must inherit the **strictest ACL of their sources**, otherwise they leak (compare Module 06 incident 2, legacy). Either build summaries per permission partition or filter edges and communities by the caller's groups before expansion. Entity name collisions across tenants ("Acme" in two customers) must be namespaced by tenant; resolution should never merge across tenants.

Evaluate with a multi-hop question set where each question lists its supporting chunks; score retrieval recall of the full evidence set and answer attribution, and compare against the hybrid baseline on the same questions. A graph that does not beat the baseline on these is not worth its index.

Cross-Connection Scenarios

- Module 06 section 2.4 (legacy) hybrid fusion is the baseline a graph must beat.
- 6.8 decomposition feeds multi-hop questions into graph or sub-query retrieval.
- Module 08 access-control patterns apply to every derived artefact.
- Module 09 agentic retrieval can call graph expansion as a tool.
- 7.7 error analysis tells you whether failures are relational at all.

Production Anti-Patterns

- Building a graph for FAQ-style questions that BM25 already solves: indexing spend with no recall gain.
- No entity resolution, so duplicate nodes fragment the graph and multi-hop paths break.
- Community summaries visible to users who cannot see the underlying documents: a permission leak through derived data.

Minimal Working Code

Entity resolution with normalised keys, token-set Jaccard and union-find, returning canonical ids and a merge report.

```
import re

SUFFIX = {"inc", "corp", "corporation", "ltd", "llc", &
quot;co"}

def norm(name):
    toks = [t for t in re.findall(r"[a-z0-9]+", name.lower()) if t not in SUFFIX]
    return frozenset(toks)

class UF:
    def __init__(self, n):
```

```
        self.p = list(range(n))
    def find(self, x):
        while self.p[x] != x:
            self.p[x] = self.p[self.p[x]]
            x = self.p[x]
        return x
    def union(self, a, b):
        self.p[self.find(a)] = self.find(b)

def resolve(names, thresh=0.6, aliases=None):
    aliases = aliases or {}
    toks = [norm(aliases.get(n.lower(), n)) for n in names]
    uf = UF(len(names))
    for i in range(len(names)):
        for j in range(i):
            inter, union = len(toks[i] & toks[j]), len(toks[i] | toks[j])
            if union and inter / union >= thresh:
                uf.union(i, j)
    groups = {}
    for i, n in enumerate(names):
        groups.setdefault(uf.find(i), []).append(n)
    canon = {n: g[0] for g in groups.values() for n in g}
    report = [g for g in groups.values() if len(g) > 1]
    return canon, report

names = ["Acme Corp.", "ACME Corporation", "Acme Robotics", "Internati
onal Business Machines", "IBM", "Globex"]
canon, report = resolve(names, aliases={"ibm": "International Business Machines"})
print(canon)
print("merged groups:", report)
assert canon["IBM"] == canon["International Business Machines"] and canon["Acme
Corp."] != canon["Acme Robotics"]
```

Neighbourhood expansion over edges that carry source documents and ACL groups; edges the caller cannot see are dropped before expansion, so hidden evidence never influences traversal.

```
from collections import deque

def visible_sources(edge, groups):
    return [d for d, acl in edge["sources"] if acl & groups]

def neighbourhood(edges, seeds, groups, hops=2):
    adj = {}
    for e in edges:
        srcs = visible_sources(e, groups)
        if srcs:                                # drop edges with no readable evidence
            adj.setdefault(e["a"], []).append((e["rel"], e["b"], srcs))
            adj.setdefault(e["b"], []).append(("~" + e["rel"], e["a&qu
ot;], srcs))
    seen, q, out = set(seeds), deque((s, 0) for s in seeds), []
    while q:
        node, d = q.popleft()
        if d == hops:
            continue
        for rel, nxt, srcs in adj.get(node, []):
            out.append((node, rel, nxt, srcs))
            if nxt not in seen:
                seen.add(nxt)
                q.append((nxt, d + 1))
    return out

edges = [
    {"a": "acme", "rel": "supplies", "b": "globex&
quot;, "sources": [("d1", {"sales"})]},
    {"a": "globex", "rel": "located_in", "b": "ger
```

```
many", "sources": [("d2", {"public"}), ("d3", {"legal&q
uot;})]},
    {"a": "globex", "rel": "under_audit_by", "b": "
;regulator", "sources": [("d4", {"legal"})]},
]
for g in ({"sales", "public"}, {"legal"}):
    print(sorted(g), [(a, r, b, s) for a, r, b, s in neighbourhood(edges, ["acme"], g)])
```

Interview Angle

Q: When would you choose GraphRAG over hybrid retrieval and what evidence do you need first? **A:** When a labelled multi-hop or corpus-wide question set shows hybrid recall of the full evidence set is poor and error analysis (7.7) attributes failures to missing relations, not to chunking or ranking. Trade-off: heavy indexing cost and staleness against recall on relational queries. Check: the graph variant beats the hybrid baseline on evidence recall and answer correctness at acceptable cost per update.

Q: How do you keep a knowledge graph fresh and permission-safe? **A:** Store source document ids and ACLs on every edge, process document changes incrementally, tombstone edges when their last source is deleted, and rebuild communities on a schedule. Filter edges by caller groups before traversal; build or tag summaries with the strictest source ACL. Check: a canary user without access must never retrieve a node, edge or summary derived from a restricted document.

Q: What does global search answer that local search cannot? **A:** Corpus-level questions (themes, trends, "what do most documents say about X") that need aggregation over many entities. Local search only expands around entities named in the query. Global search costs more (map-reduce over summaries) and can be coarse, so route only aggregate questions to it.

6.7 Embedding Models in Practice: Selection, Fine-Tuning, Matryoshka and Internal Benchmarks

Conceptual Deep Dive

Select an embedding model with your own retrieval benchmark: real (or realistically synthesised) queries, labelled relevant chunks, and recall@k and nDCG measured on your corpus. Public leaderboards (MTEB is the common reference) are a shortlist tool; domain, language mix, query style and document length differ from your data, and a model ranked first overall can lose on your corpus.

Cost drivers are dimension, precision and input length. **Matryoshka**-trained models are trained so that the leading dimensions carry most of the signal, so you can truncate vectors (for example to a quarter of the dimensions) and re-normalise, trading some recall for memory and latency. Binary or int8 quantised embeddings reduce memory by 4-32x; a standard pattern is coarse search on compact codes followed by float rescoring of the top candidates (extends Module 06 section 1.4, legacy). Always measure recall@k against full-precision exact search on your data rather than trusting a vendor claim.

Conventions matter as much as architecture: many models want different prefixes or instructions for queries and documents, a specific normalisation, and have a maximum input length beyond which text is silently truncated. Check the model card; a missing query prefix can cost double-digit recall points. Multilingual coverage must be tested per language, not assumed. Documents longer than the limit need chunking (an edge case that silently drops the tail); domain jargon and code identifiers are tokenised poorly, so add sparse retrieval (BM25) alongside.

Fine-tuning uses contrastive training: queries paired with positive chunks, in-batch negatives plus mined **hard negatives** (retrieved by the current model but not relevant), from synthetic queries generated by an LLM or from logged queries with click or citation labels. Beware false negatives among mined negatives, and train/test leakage: if evaluation queries are generated from the same chunks used for training, scores inflate. Judge success by recall@k uplift on a held-out, human-curated set, not by training loss. Fine-tuning is worth it when a gap remains after fixing chunking, hybrid search and reranking, and you have thousands of in-domain pairs.

Versioning: the embedding model id, dimension, prefix and normalisation are part of the index identity. Vectors from two models (or two versions) are not comparable; migrate with a new collection and an alias flip (Module 06 section 3.10, legacy), re-embedding the whole corpus, never mixing versions in one index. Operationally, size the embedding service by batch throughput on GPU or CPU, and budget the re-embed: corpus tokens divided by throughput, plus the rate limits of any hosted API.

Cross-Connection Scenarios

- Module 06 section 1.4 (legacy) covers quantisation of stored vectors.

- Module 06 section 3.10 (legacy) is the blue/green index migration used for model changes.
- 7.6 pairwise judging can help compare retrieval outputs when labels are scarce.
- 12.6 Drill 2 estimates the cost and wall-clock of re-embedding.
- 14.1 golden-set practice applies to the retrieval benchmark.

Production Anti-Patterns

- Choosing the top MTEB model without testing on your queries: leaderboard rank does not transfer to your domain.
- Mixing embeddings from two model versions in one index: distances become meaningless and recall silently drops.
- Fine-tuning on queries generated from the same chunks used for evaluation: leakage inflates gains that vanish in production.

Minimal Working Code

Simulated Matryoshka-style truncation on synthetic clustered vectors (variance decays across dimensions to mimic front-loaded information), plus binary quantisation with float rescoring of the top 100. Recall is measured against full-dimension exact top-10.

```
import numpy as np

rng = np.random.default_rng(0)
n_docs, d, n_q = 4000, 256, 200
scale = 1.0 / np.sqrt(1 + np.arange(d) / 8.0)
centres = rng.normal(size=(40, d)) * scale
X = centres[rng.integers(0, 40, n_docs)] + 0.5 * rng.normal(size=(n_docs, d)) * scale
Q = X[rng.choice(n_docs, n_q, replace=False)] + 0.2 * rng.normal(size=(n_q, d)) * scale
unit = lambda A: A / np.linalg.norm(A, axis=1, keepdims=True)
X, Q = unit(X), unit(Q)

topk = lambda S, k: np.argsort(-S, axis=1)[:, :k]
recall = lambda found, truth: np.mean([len(set(f) & set(t)) / len(t) for f, t in zip(found, truth)])
truth = topk(Q @ X.T, 10)

for dim in (256, 128, 64, 32):
    Xt, Qt = unit(X[:, :dim]), unit(Q[:, :dim])
    print("dims", dim, "recall@10", round(recall(topk(Qt @ Xt.T, 10), truth), 3))

ham = np.stack([(X > 0) != (q > 0) for q in Q]).sum(2)      # Hamming distance on sign bits
cand = np.argsort(ham, axis=1)[:, :100]
rescored = [c[np.argsort(-(X[c] @ q))[:10]] for c, q in zip(cand, Q)]
print("binary top-10:", round(recall(np.argsort(ham, axis=1)[:, :10], truth), 3),
      "| binary top-100 + float rescore:", round(recall(rescored, truth), 3))
```

InfoNCE with in-batch negatives, then the same training with mined hard negatives; the toy table has decoy documents near each query, so low in-batch loss does not imply good retrieval.

```
import numpy as np

rng = np.random.default_rng(0)
nq, nd, d, tau = 64, 300, 16, 0.5
Q = rng.normal(size=(nq, d))
D0 = np.vstack([Q, Q[rng.integers(0, nq, nd - nq)] + 0.6 * rng.normal(size=(nd - nq, d))])
D0 = D0 + rng.normal(size=D0.shape)                    # docs 0..nq-1 are the positives, rest are deco
ys

def loss_grad(D, hard):
    cand = np.concatenate([np.arange(nq), hard])
    S = Q @ D[cand].T / tau
    P = np.exp(S - S.max(1, keepdims=True)); P /= P.sum(1, keepdims=True)
    loss = -np.log(P[np.arange(nq), np.arange(nq)]).mean()
    Y = np.zeros_like(P); Y[np.arange(nq), np.arange(nq)] = 1
    G = np.zeros_like(D); np.add.at(G, cand, (P - Y).T @ Q / (tau * nq))
    return loss, G

recall1 = lambda D: (np.argmax(Q @ D.T, 1) == np.arange(nq)).mean()
```

```
for use_hard in (False, True):
    D = D0.copy()
    for _ in range(200):
        hard = np.array([], dtype=int)
        if use_hard:
            S = Q @ D.T
            S[np.arange(nq), np.arange(nq)] = -1e9        # exclude the positive, mine the best decoy
            hard = np.unique(np.argmax(S, 1))
        loss, G = loss_grad(D, hard)
        D -= 0.5 * G
    print("hard negs" if use_hard else "in-batch ", "loss", round(loss, 3),
 "recall@1 before/after", round(recall1(D0), 2), round(recall1(D), 2))
```

Interview Angle

Q: How do you pick an embedding model for a multilingual support corpus? **A:** Shortlist from MTEB-style multilingual results, then build a benchmark of real queries per language with labelled answers and measure recall@k, nDCG, latency and memory per candidate; test code-switching and long documents. Trade-off: larger models and dimensions versus cost; Matryoshka truncation can recover memory. Check: per-language recall within a tolerance of the best language, and a BM25 hybrid baseline for jargon.

Q: Recall dropped after a "better" embedding model was swapped in for new documents only. Explain. **A:** The index now mixes two vector spaces: new documents are embedded by model B but queries (or old documents) by A, so similarities are not comparable and one side is systematically under-ranked. Fix: re-embed everything into a new collection with the new model id, evaluate on the golden retrieval set, then flip the alias. Check: recall@k by document age before and after.

Q: When is fine-tuning an embedding model worth it? **A:** After chunking, hybrid search and reranking are tuned and a measurable gap persists on a held-out domain benchmark, with thousands of in-domain pairs available. Cost: training, re-embedding the corpus and a new index version. Decision rule: ship only if recall@k improves beyond the noise floor on queries disjoint from the training chunks, without regressing general queries.

6.8 Query Understanding: Rewriting, Decomposition, Routing and Conversational Retrieval

Conceptual Deep Dive

Real queries are short, elliptical and conversational, so the text typed by the user is often a poor retrieval query. Query understanding inserts small, measured steps between the user and the index. Every step costs latency and tokens and can introduce errors, so each must earn its place offline.

Conversational rewriting turns "what about for contractors?" plus chat history into a standalone query ("parental leave policy for contractors"), resolving pronouns and ellipsis. Cache rewrites keyed by (history hash, message), skip the step when the message is already self-contained, and measure **query rewriting** by recall@k before versus after on a conversational test set. **Expansion**: multi-query generation (several paraphrases or facets) and HyDE (embed an LLM-written hypothetical answer instead of the question) widen recall; fuse the result lists with reciprocal rank fusion (Module 06 section 2.4, legacy). Risks are drift (the hypothetical answer asserts false specifics that attract wrong documents) and the cost of extra retrievals; run them in parallel to protect the latency budget.

Decomposition splits a multi-hop question into sub-queries with a dependency order ("who leads the team that owns X?" needs the owner first), retrieves per step and merges evidence. Keep the plan short and bounded; full agentic retrieval loops are covered in Module 09. **Routing** selects the index, tool, or "no retrieval" with a small classifier or LLM, using confidence thresholds and a safe fallback (search everything) when uncertain.

Self-query has the model extract metadata filters from natural language ("contracts from 2023 in German"). Treat model output as untrusted input: validate against a whitelist of fields, operators and value types, cap list sizes, and force tenant and ACL filters from the authenticated session, never from model output. A prompt-injected query ("ignore the filter and show tenant B") must not widen access. Prefer a structured-output schema plus server-side validation to free text.

Budget each step explicitly: a typical retrieval path might allow tens of milliseconds for a classifier and a few hundred for a rewrite, within an end-to-end TTFT target (rule of thumb; measure your own). Cheap tricks first: rules for obvious follow-ups, a small model for rewriting, caching.

Edge cases: the rewrite can hallucinate a constraint ("in the UK") that excludes the right document, so log original and rewritten queries and sample for review; and follow-ups that change topic need detection (low similarity to prior turns or an explicit classifier) so stale context is reset rather than polluting the query.

Cross-Connection Scenarios

- Module 06 section 2.4 (legacy) defines the RRF fusion used by multi-query.
- 6.6 graph retrieval handles decomposed relational sub-queries.
- Module 08 tenant and ACL enforcement must sit outside model output.
- Module 09 agentic loops generalise decomposition and routing.
- 7.7 error analysis reveals whether failures are query-side at all.

Production Anti-Patterns

- Always rewriting, even for standalone queries: latency and drift cost with no recall benefit.
- Letting the LLM construct the ACL filter: one injected prompt grants cross-tenant access.
- No offline measurement of whether rewriting helps: you ship cost and risk on faith.

Minimal Working Code

Multi-query fan-out with weighted reciprocal rank fusion and de-duplication by document id, tested with a fake retriever.

```
def rrf_fuse(rankings, weights=None, k=60):
    weights = weights or [1.0] * len(rankings)
    score, best_rank = {}, {}
    for w, ranking in zip(weights, rankings):
        seen = set()
        for rank, doc in enumerate(ranking):
            if doc in seen:                    # de-duplicate within one list
                continue
            seen.add(doc)
            score[doc] = score.get(doc, 0.0) + w / (k + rank + 1)
            best_rank[doc] = min(best_rank.get(doc, rank), rank)
    return sorted(score, key=lambda d: (-score[d], best_rank[d], d))

def multi_query(queries, retrieve, weights=None, top_n=5):
    return rrf_fuse([retrieve(q) for q in queries], weights)[:top_n]

FAKE = {
    "leave policy contractors": ["d3", "d1", "d9"],
    "contractor parental leave": ["d1", "d3", "d4"],
    "time off for contingent workers": ["d7", "d1", "d1"],
}
out = multi_query(list(FAKE), FAKE.__getitem__, weights=[1.0, 1.0, 0.5])
print(out)
assert out[0] == "d1" and len(out) == len(set(out))
```

A self-query filter validator: allow-listed fields, operators and value types; the tenant filter is injected from the session; everything else is rejected, including an injection attempt.

```
ALLOWED = {
    "lang": ({"eq", "in"}, str),
    "year": ({"eq", "gte", "lte"}, int),
    "doc_type": ({"eq", "in"}, str),
}

def validate(model_filters, tenant_id, max_list=10):
    clean = []
    if not isinstance(model_filters, list) or len(model_filters) > 5:
        raise ValueError("filters must be a short list")
    for f in model_filters:
        if not isinstance(f, dict) or set(f) != {"field", "op", "value"}:
            raise ValueError("bad filter shape")
```

```
        field, op, val = f["field"], f["op"], f["value"]
        if field not in ALLOWED:
            raise ValueError(f"field not allowed: {field}")
        ops, typ = ALLOWED[field]
        if op not in ops:
            raise ValueError(f"operator not allowed: {op}")
        vals = val if op == "in" else [val]
        if op == "in" and (not isinstance(val, list) or len(val) > max_list):
            raise ValueError("bad list value")
        if any(isinstance(v, bool) or not isinstance(v, typ) for v in vals):
            raise ValueError("bad value type")
        clean.append({"field": field, "op": op, "value": val})
    clean.append({"field": "tenant_id", "op": "eq", "value&q
uot;: tenant_id})   # from the session, never the model
    return clean

print(validate([{"field": "year", "op": "gte", "value":
 2023}], "tenant-a"))
for bad in ([{"field": "tenant_id", "op": "eq", "value"
: "tenant-b"}],
            [{"field": "year", "op": "gte", "value": &q
uot;2023; DROP"}],
            [{"field": "lang", "op": "regex", "value":
".*"}]):
    try:
        validate(bad, "tenant-a")
        raise SystemExit("injection accepted")
    except ValueError as e:
        print("rejected:", e)
```

Interview Angle

Q: Users ask "what about for contractors?" as a follow-up. Design the retrieval path. **A:** Detect that the message depends on history, rewrite it to a standalone query with a small model using the last few turns, retrieve with hybrid search (optionally multi-query fused with RRF), and answer; if the topic changed, reset context. Keep the original and rewritten queries in the trace. Trade-off: extra 100s of ms versus recall. Check: recall@k on a follow-up test set with and without rewriting.

Q: How do you prevent a model-written metadata filter from becoming a security hole? **A:** Treat it as untrusted input: structured output, validation against an allow-list of fields, operators and types, size limits, and rejection (not repair) of anything else. Tenant and ACL predicates come from the authenticated session and are appended server-side, ideally enforced again in the datastore. Check: a red-team suite of injection prompts must never retrieve another tenant's documents.

Q: How would you measure whether query rewriting is worth its latency? **A:** Offline, compare recall@k and answer correctness on a conversational golden set with rewriting on and off, sliced by standalone versus follow-up queries. Online, A/B the p95 TTFT and resolution rate. Keep it if the uplift on follow-ups exceeds the noise floor and latency stays within budget; restrict it to queries a cheap classifier flags as dependent.